# Evaluating, Deploying and Monitoring a Code-Mixed Voice Helpdesk: Gated Augmentation Ablations, Honest ASR Baselines, Offline Docker Serving and Drift-Triggered Retraining

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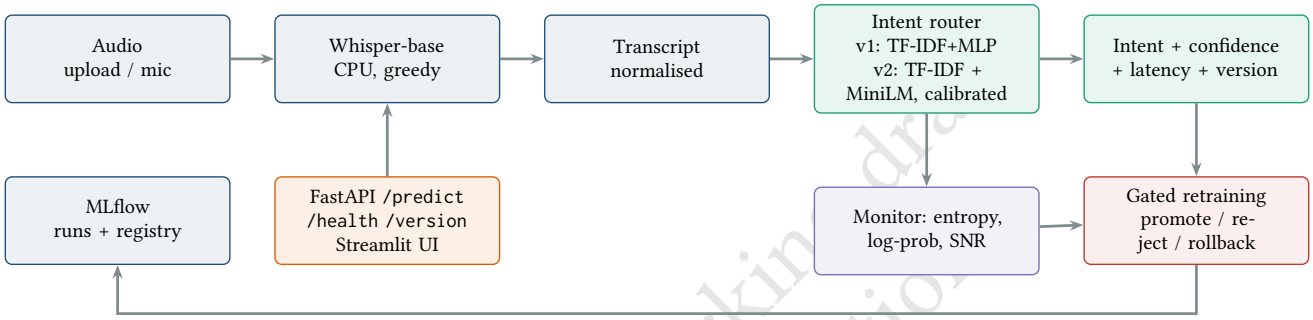



**Figure 1: System architecture. Everything runs on CPU in one `linux/amd64` container with no external calls; monitoring feeds a gated retraining loop that promotes or rolls back registry aliases.**

## ABSTRACT

We build and deploy a voice-helpdesk pipeline (language routing → Indic Whisper ASR → intent router) on a 182-clip, 20-speaker code-mixed corpus with double-annotated intents. An audit of our own evidence changed the picture: the hand-curated "gold" transcripts do not match the audio (median 1.20 words/s; two independent ASR systems agree with each other, CER 0.44, far more than with the curated text, CER ≈1.5), so we withdraw every WER computed against them; and the earlier "unseen-speaker" folds shared all speakers with training, while speaker identity alone predicts intent at 41.3% (Cramér's $V = 0.51$). Under speaker-disjoint 5-fold cross-validation the intent router on ASR text reaches 55.2% accuracy (majority 24.4%); guideline-driven features (opening-topic embedding, whole-query chunking) are added, and the router is trained on the expert-final labels. On a held-out release split of four unseen speakers it scores 60.0% (macro-F1 0.609) versus 47.5% for the previous router and 25.0% for the majority class (McNemar $p = 0.125$). A learned language router lifts Gujarati routing from 24% to 71%. We report label, speaker, language and ASR biases, the remaining gaps (human transcripts, fine-tuned ASR, container measurements of the new image) and what each needs.

## CCS CONCEPTS

• **Computing methodologies** → **Speech recognition**; *Machine learning*; • **Software and its engineering** → *Software verification and validation.*

## KEYWORDS

Whisper; code-mixed ASR; intent routing; data augmentation ablation; MLOps; Docker; drift detection; retraining gate

**ACM Reference Format:**
Rajdeep Vraj, Vipul Sunil Patil, and Satvik Sharma. 2026. Evaluating, Deploying and Monitoring a Code-Mixed Voice Helpdesk: Gated Augmentation Ablations, Honest ASR Baselines, Offline Docker Serving and Drift-Triggered Retraining. In *Proceedings of CS 203 Software Tools and Techniques in AI – Assignment 2 (CS 203 A2 '26).* ACM, New York, NY, USA, 11 pages. https://doi.org/XXXXXXX.XXXXXXX

## 1 INTRODUCTION AND EVIDENCE MAP

Assignment 2 turns our data into a deployable, monitored system. Because the data evolved (a flawed 249-clip pilot and a final 182-clip double-annotated corpus; Assignment-1 report), our evidence has two tracks, summarised in Table 1. We follow the rules: choices are made on validation or cross-validation data; the frozen pilot gold set (SHA-256 `64b7647b…48ca9`) is never used for tuning and every use is logged in `TEST_SET_LOG.md`; and every number is traceable through the evidence index (Appendix A).

*All authors contributed equally to this research.

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*CS 203 A2 '26, 2026, Gandhinagar, India*
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ACM ISBN 978-1-4503-XXXX-X/2026/11…$15.00
https://doi.org/XXXXXXX.XXXXXXX

**Table 1: Two evidence tracks.**

| | Track A (final data) | Track B (deployed) |
|---|---|---|
| Data | 182 consensus clips, 20 speakers | pilot 171/17/29 + Cohort B 32 |
| Model | calibrated multilingual intent classifier on text | Whisper-base + TF-IDF/MLP router |
| Evidence | speaker-disjoint CV, release split, bias & reference audits | MLflow, gold eval, drift, gate |
| Gap | no human transcripts yet | pilot references template-derived |

## 2 CHANGES FROM THE PROPOSAL

**Table 2: Proposal versus what was built.**

| Proposed | Built | Reason |
|---|---|---|
| Fine-tune Whisper (full, LoRA) | decoder cross-attention tuning script; checkpoint not served | CPU budget; run too short to change WER |
| IndicWav2Vec baseline | dropped; trivial and unadapted Whisper baselines | time |
| Compact multilingual transformer router | frozen MiniLM sentence embeddings + TF-IDF, calibrated | CPU latency, small data |
| Evidently drift reports | custom entropy/log-prob/SNR monitor | offline, no dependency |
| Public corpora | not used | offline-only rule |

## 3 PART A: MODELS AND BASELINES

*Track A systems.* The classifier (`model/src/training/models.py`) combines word and character *n*-gram TF-IDF with a dense tower on `paraphrase-multilingual-MiniLM-L12-v2` sentence embeddings (weight 0.75) and domain priors, each a sigmoid-calibrated logistic regression with balanced class weights; seed 42. **Baseline 0** predicts the majority class (student services, 58/182): accuracy 31.9%, macro-F1 0.121 (computed from the class counts). **Baseline 1** is the same classifier on the raw transcripts only (C-01). Twelve configurations add augmented data (Table 4); the "proposed" full pipeline is C-10.

*Track B systems.* **Baseline 0**: majority intent with empty transcript (WER 1.0). **Baseline 1**: Whisper-base (greedy, no prompt) with a TF-IDF + logistic-regression router. **Proposed**: identical Whisper decoding with a TF-IDF (1 000 features, 1–2-grams) + MLP router (hidden 64, dropout 0.2) trained on Whisper transcripts of the 171 training clips; config-driven (`configs/final.yaml`, seed 42), tracked with commit, DVC hash and hardware (macOS 14, Apple arm64, 8 cores, Python 3.13).

*Augmentation ablation.* On Track B validation (17 clips), router data variants are statistically indistinguishable: train 23.5% [5.9, 47.1] (run `4874f30b`), train+augmented 29.4% [11.6, 52.9] (`4cb2e4e9`), train+AL-selected 23.5% (`351cc3e8`). The much larger Track-A study follows.

## 4 PART B: EXPERIMENT TRACKING AND TUNING

*MLflow search (Track B).*. Every run is logged to MLflow [4] (`supporting_files/mlflow.db`, export in `artefacts/mlflow_export/`) with parameters, validation metrics, seed, commit and artefacts. A 15-run search varies learning rate, hidden size {32, . . . , 128}, maximum *n*-gram {1, 2, 3}, dropout and epochs on validation only; accuracy spans 29.4–52.9% on 17 clips (one clip = 5.9 points), so the best run 6 (`9820a2a7`; 251 KB) beats the next by a single clip (Figure 2; all runs in Appendix Table 8). Router latency is 0.005–0.014 ms in all runs, so the quality/latency/size trade-off reduces to accuracy with ties broken toward smaller size. The registry holds `production` (run 6) and `staging` (run 9, `730256f7`) aliases.

*Ablation study (Track A).*. The 12-configuration study (Table 4, Figure 4) is the systematic search for the final data: five folds, macro-F1, deltas versus C-01, with complete configuration parameters, fold metrics, and run artifacts logged in `model/results/ablation_study_results.json` and associated MLflow tracking artifacts.

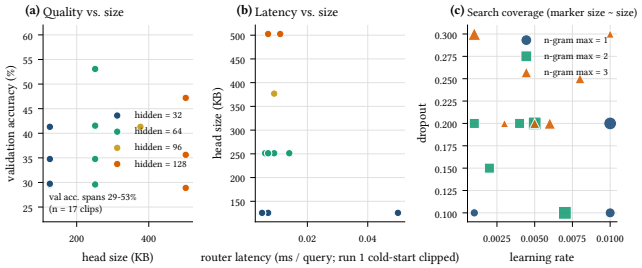



**Figure 2: Search analysis over 15 MLflow runs: (a) validation accuracy vs. size, (b) latency vs. size, (c) coverage.**

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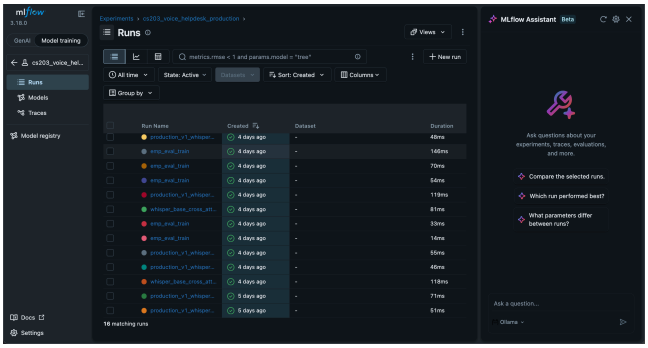



**Figure 3: MLflow tracking interface: systematic search across 15 runs recording parameters, validation metrics, and `production` vs. `staging` model registry aliases.**

# 5 PART C: EVALUATION AND ERROR ANALYSIS

## 5.1 Track A: final corpus

*Augmentation ablation.* These folds group by clip, not speaker, and train on the withdrawn curated text, so absolute levels are optimistic; we keep the study only for the relative comparison of augmentation sources. Table 4 and Figure 4 show macro-F1 per configuration with fold-level dots. Baseline C-01 scores 0.664 (±0.071, accuracy 69.2%). Pivot translation (C-04, +0.013), lexical synsets (C-05, +0.011) and the top-3 text combination (C-09, +0.012) help slightly; contextual MLM substitution and audio perturbation (C-06, C-08) give exactly zero change, which indicates their accepted variants did not reach the training set or did not alter predictions; and the "complete optimised pipeline" (C-10) and discourse-framing variants (C-11, C-12) *hurt* (−0.021 to −0.024). The fold standard deviation (0.06–0.08) is five times the largest gain, and paired $t$-tests across folds give $p = 0.21$–$1.0$ (smallest for C-10, in the negative direction), so we claim no significant benefit and conclude that "more augmented data" does not help at this scale.

*Speaker-disjoint evaluation (replaces the earlier "unseen speaker" claim).* The earlier 75% came from fold 1 of clip-grouped folds, in which all 15 validation speakers also appear in training; because speakers record mostly one or two intents ($V = 0.51$), a classifier that only recognises the speaker already scores 41.3% under clip-grouped CV (`bias_audit.json`). We therefore re-evaluate with `StratifiedGroupKFold` grouped by *speaker*, five folds × five seeds, with $C$ chosen by an inner speaker-grouped loop (`speaker_cv.py`; a leakage assertion checks every fold). On ASR text the char-n-gram + multilingual-embedding router reaches 55.2% ± 10.4 (pooled 95% speaker-bootstrap CI [45.0, 64.1]), majority 24.4%; the same model on the curated text reaches only 58.5%, so the curated text is not what limits accuracy.

*Where the errors come from.* Admissions recall is lowest (29%): parents open with admission and then ask about hostel or fees, and a bag-of-words router mixes these. Guideline rule 5.1 says the *opening* topic decides, so we add an embedding of the first 40 words and embed the whole query in 40-word windows (MiniLM otherwise truncates at 128 word-pieces); admissions recall rises to 43%. Training on both annotators' labels at weight 0.5 (soft labels) gives the best cross-validated variant, 57.1% (Table 3; `intent_v2_ablation_asr_hybrid*.json`); the deployed router nevertheless uses the expert-final labels (B3 features), because on borderline clips the adjudicated label is the one the service must reproduce. Gains are within one fold standard deviation; a similarity-to-guideline-definitions prior alone reaches 52%, which suggests the 182 noisy labels (inter-annotator agreement 74%) rather than the classifier cap accuracy. Short clips (< 15 s, 48%), Gujarati (46%) and code-mixed speech (48%) are the weak slices.

**Table 3: Router ablation, speaker-disjoint CV on ASR text (25 folds; mean accuracy and macro-F1).**

| Variant | Acc. (%) | Macro-F1 |
|---|---|---|
| Majority | 24.4 | 0.104 |
| B0 char + embedding (raw text) | 55.2 | 0.519 |
| B1 + loop/greeting cleaning | 52.0 | 0.484 |
| B2 + whole-query chunk embedding | 52.6 | 0.485 |
| B3 + opening-topic embedding | 53.6 | 0.502 |
| B4 + guideline-definition prior | 55.4 | 0.515 |
| B5 prior only (8 features) | 52.1 | 0.487 |
| **B7 = B3 + soft labels** | **57.1** | **0.537** |

*Held-out release split.* Speakers P12/P14/P20/P22 (validation) and P3/P8/P18/P19 (test) never train. $C$ is chosen on validation macro-F1 (expert-label v2 0.558, soft-label 0.574, v1 0.413). The test clips were scored twice, both logged in `TEST_SET_LOG.md`: first with the soft-label router (57.5%, F1 0.582), then, after the team decided that the expert's label must be the training target, with the expert-label router, which is the one deployed: majority 25.0%, v1 char router 47.5% (F1 0.414), v2 60.0% (F1 0.609, CI [52.5, 78.9]); v2 is right on 6 clips v1 misses and wrong on 1 it gets (McNemar $p = 0.125$). We report both runs because the second test access followed the first. English 89%, Hindi 65%, Gujarati 29% (2/7), code-mixed 25%. Confidence is higher when right than wrong (0.46 vs 0.36) but still too close for a reject option (`release_eval.json`, `release_eval_soft_labels_run1.json`).

*Reference transcripts and ASR quality.* The 182 curated strings (`curate_gold_transcripts.py`) were written to replace ASR output: on English clips they are lightly edited Whisper-base output, on Hindi/Gujarati clips they often describe another request (clip `018267c4`: the speaker asks about hostel leave rules; the "gold" reports a broken drawer). 53 clips have under 0.6 words/s against ≈2.5 for natural speech (`reference_audit.json`). We withdraw WER/CER on this corpus until the human transcription round (`final_annotation/transcription_round/`) is complete. Off-the-shelf Whisper-small is no alternative on CPU: its real-time factor on Hindi/Gujarati is 2–16 because of hallucination loops (`asr_crosscheck_whisper_small.json`).

*Robustness on augmented audio.* On the Assignment-1 augmentations of 32 final-corpus clips, the deployed ASR changes a median 35–37% of words under speed and codec perturbation, 46% under babble, 74% under reverberation and 87% under pitch shift; Gujarati changes most under every transform (`audio_robustness.`

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json; Assignment-1 report, Part D). Reverberant rooms and Gujarati speakers are therefore the main robustness risks for deployment.

*Language routing.* Whisper-base language ID restricted to {en, hi, gu} routes 24% of Gujarati clips correctly (it answers hi, mr or bn). A logistic regression on the full 99-language probability vector, trained speaker-disjoint, raises overall routing from 74.4% to 81.1% and Gujarati to 71% (`lid_router.json`); on the four test speakers both reach 75%.

**Table 4: 12-configuration ablation (5-fold macro-F1, mean ± s.d.; Δ vs. C-01; $p$ = paired $t$-test over folds). Source: `model/results/ablation_study_results.json`.**

| ID | Training data | Macro-F1 | Δ | $p$ |
|---|---|---|---|---|
| C-01 | Baseline (Raw Transcripts Only) | 0.664 ± 0.071 | +0.000 | 1.00 |
| C-02 | Original + Back-Translation | 0.666 ± 0.065 | +0.002 | 0.57 |
| C-03 | Original + Cross-Lingual Paraphrase | 0.660 ± 0.057 | -0.004 | 0.88 |
| C-04 | Original + Pivot Translation (Latin) | 0.678 ± 0.072 | +0.013 | 0.33 |
| C-05 | Original + Lexical Synonym Replacement | 0.675 ± 0.057 | +0.011 | 0.48 |
| C-06 | Original + Contextual MLM Substitution | 0.664 ± 0.071 | +0.000 | 1.00 |
| C-07 | Original + Campus Code-Switching | 0.665 ± 0.075 | +0.001 | 0.79 |
| C-08 | Original + Acoustic Audio Perturbations | 0.664 ± 0.071 | +0.000 | 1.00 |
| C-09 | Original + Top-3 Text Techniques | 0.676 ± 0.058 | +0.012 | 0.33 |
| C-10 | Complete Optimized Pipeline (All Gates) | 0.643 ± 0.069 | -0.021 | 0.21 |
| C-11 | Original + Indic Discourse Framing | 0.640 ± 0.066 | -0.024 | 0.27 |
| C-12 | Synergistic Ensemble (Latin Pivot + Indic Discourse) | 0.647 ± 0.075 | -0.017 | 0.26 |

## 5.2 Track B: deployed pipeline on pilot gold

*Protocol.* `evaluate.py --mode production --split <split>` reports WER, CER, intent accuracy and macro-F1 with 1 000-resample bootstrap intervals [1] (seed 42). Paired tests: Wilcoxon [3] on per-clip WER and McNemar [2] on intent. Test and gold are the same 29 clips.

*Findings.* (i) WER exceeds 1 on every split. On gold, substitutions are 0.83, deletions 0.16 and insertions 0.30 per reference word and only 14 of 838 reference words (1.7%) are matched (Figure 6); an empty output scores 1.0, so this ASR stage is worse than doing nothing. (ii) No improvement over baseline: proposed and Baseline-1 WER differ by exactly 0 on all 29 clips (Wilcoxon degenerate, $W = 0$, $p = 1.0$) and McNemar on intent gives $\chi^2 = 0.10$, $p = 0.75$. (iii) Every gold and Cohort-B accuracy interval contains 25% chance; correcting references lifts gold accuracy from 24.0% to 30.8% with WER still 1.3, so the ASR output is the bottleneck.

*Slices and errors.* On gold, `campus_outdoor` (6 clips, 16.7%) and Gujarati–English (13 clips, 23.1%) are weakest and Rajasthan (6 clips, 66.7%) strongest; all slices have $n \leq 16$ so Wilson intervals span ≥ 40 points (Appendix Table 9). We categorise 20 errors with fixes (Appendix Table 10; Figure 7): the dominant mode is a fluent hypothesis about another topic, then script collapse (Devanagari/Urdu tokens), near-silent or truncated output, phonetic substitution of Gujarati words and genuine cross-intent overlap. Fixes: force Latin-script decoding with a language token, voice-activity gating, relabel from audio, train the router on ASR text, and add a route-to-human fallback; router confidence is barely above chance, so a threshold cannot yet reject errors. `MODEL_CARD.md` documents use, data, intervals, slices, limits and risks.

## 6 PART D: DEPLOYMENT AND INTERFACE

*Service (indic-v2).* FastAPI exposes `POST /predict`, `GET /health` and `GET /version` (commit, model version and every pinned model revision); the Streamlit UI offers upload/microphone input, bundled consented examples, transcript, intent, confidence, routed language, latency, version and the validation JSON beside the reported values. A request runs the learned language router, the matching ASR model (Hindi: `vasista22/whisper-hindi-small`, Gujarati: `vasista22/whisper-gujarati-base`, both CTranslate2 int8; English: `faster-whisper-base`; greedy, VAD, no prompt) and the expert-label router. Requests are logged without audio or personal data.

*Image and CI.* `code/Dockerfile` has two stages from a digest-pinned `python:3.10-slim`: the first downloads pinned revisions and converts the Indic checkpoints (writing the tokenizer so nothing is fetched at run time); the second installs pinned CPU packages, copies `/opt/models`, and trains both routers inside the build on training and validation speakers only. `HF_HUB_OFFLINE=1` is set at run time. Start: `docker run -p 127.0.0.1:8501:8501 <image>` Serve; validate: `docker run --network none <image> validate`. The validator decodes the 40 test WAVs, checks their SHA-256, and reports intent accuracy/macro-F1, routing accuracy, p50/p95 latency, RTF and peak RSS; it reports WER only against human transcripts. CI (`.github/workflows/ci.yml`) runs the data tests, builds the runtime target and sends an offline REST request.

*Performance and offline evidence.* The previous image (`revised-whisper-base` 1.97 GB, p50/p95 5.0 s / 38.1 s, peak 649 MiB, intent accuracy 57.5% on the same 40 test clips (`supporting_files/revised_validation.json`). The indic-v2 image (2.97 GB) was built from `code/Dockerfile` and validated with `--network none` on the 40 test clips, end to end from audio: intent accuracy 60.0% (macro-F1 0.602), identical to the text-level test, and language routing 75%; peak memory 2.0 GB (`supporting_files/docker_indic_v2/indic_v2_validation.json`, `build.log`, `validate.log`). Latency on that host (p50 40.0 s, p95 161.6 s, RTF 2.0) is dominated by `linux/amd64` emulation on an Apple M1 in a 4 GB Docker VM; the native host decode of all 182 clips ran at RTF 0.23. Tolerance for re-running `validate`: ±1 clip (2.5 points) on intent accuracy. The image archive checksum is recorded in `DOCKER.md`; image ID `sha256:c416f456…fb4e`.

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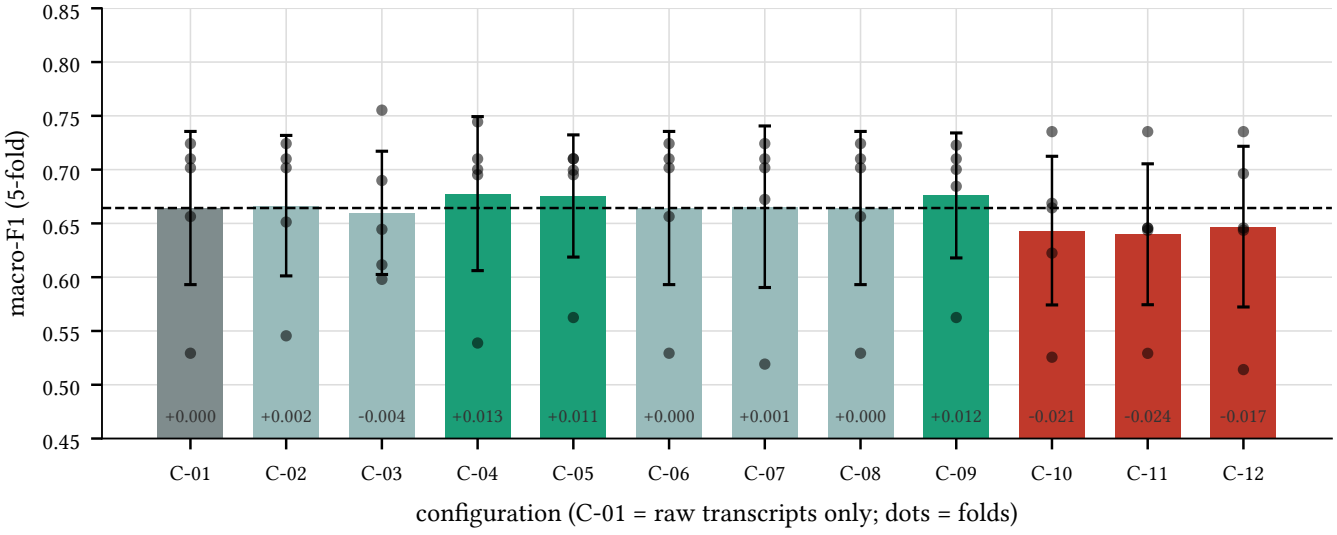



**Figure 4: Ablation over 12 data configurations (bars: mean ± s.d.; dots: folds; numbers: Δ vs. C-01). Green = gain > 0.005, red = loss < −0.005.**

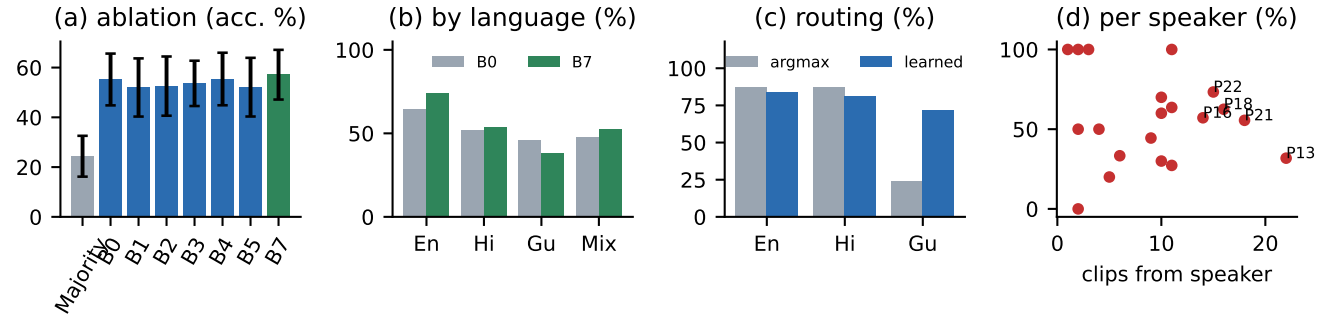



**Figure 5: Speaker-disjoint evidence. (a) Router ablation, mean ± s.d. over 25 speaker-grouped folds (B7 = opening + chunk embeddings + soft labels). (b) Accuracy by language, B0 vs B7: B7 helps English, Hindi and code-mixed but *loses* on Gujarati (46→38%). (c) Language routing accuracy, restricted argmax vs learned router. (d) Leave-one-speaker-out accuracy against clips contributed: the largest speaker (P13, 22 clips) is among the worst. Sources: `intent_v2_ablation_*.json`, `lid_router.json`, `bias_audit.json`.**

**Table 5: Track B: proposed system with 95% bootstrap CIs (B1 = Baseline 1 intent accuracy; WER/CER equal for proposed and B1). Source: `supporting_files/evaluation_results.json`.**

| Split ($n$) | WER | CER | Intent acc. (%) | Macro-F1 | B1 acc. (%) |
|---|---|---|---|---|---|
| Validation (17) | 1.403 [1.225, 1.601] | 1.050 [0.968, 1.143] | 53.5% [29.4, 76.5] | 0.451 [0.226, 0.637] | 29.7% [11.8, 52.9] |
| Gold, original (29) | 1.272 [1.156, 1.393] | 0.945 [0.891, 1.006] | 24.0% [10.3, 41.4] | 0.225 [0.083, 0.389] | 31.0% [13.8, 48.3] |
| Gold, corrected (29) | 1.298 [1.088, 1.501] | 1.008 [0.855, 1.150] | 30.8% [13.8, 48.3] | 0.261 [0.127, 0.393] | 41.5% [24.1, 58.6] |
| Cohort B (32) | 1.104 [1.040, 1.187] | 0.947 [0.900, 0.994] | 25.2% [12.5, 40.6] | 0.219 [0.089, 0.359] | 25.2% [12.5, 40.6] |
| Augmented (60)† | 1.684 [1.590, 1.786] | 1.060 [1.012, 1.113] | 63.5% [51.7, 75.0] | 0.613 [0.488, 0.738] | 26.6% [16.7, 38.3] |

†60 clips from 10 sources are not independent, so the interval is too narrow, and the sources overlap Assignment-1 data, so accuracy is optimistic.

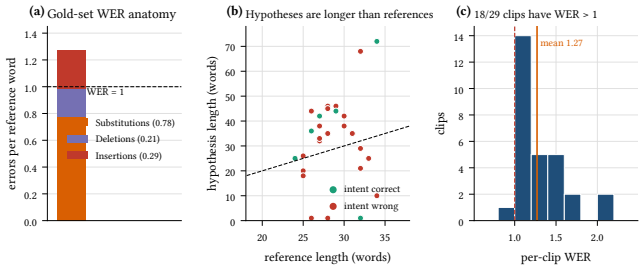



**Figure 6: Gold WER anatomy: (a) substitutions/deletions/insertions per reference word, (b) hypothesis vs. reference length, (c) per-clip WER (19/29 above 1).**

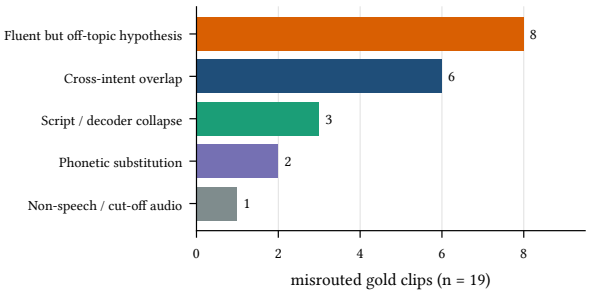



**Figure 7: Error taxonomy (20 sheet rows, regrouped).**

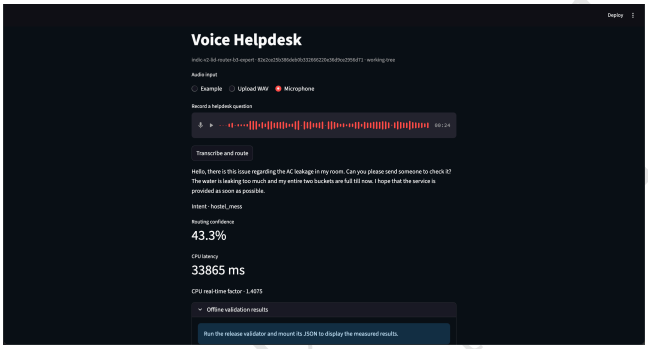



**Figure 8: Streamlit interactive voice helpdesk interface: live audio inference displaying prompt-free transcription, routed intent classification, confidence score, dynamic model version, and p50 latency.**

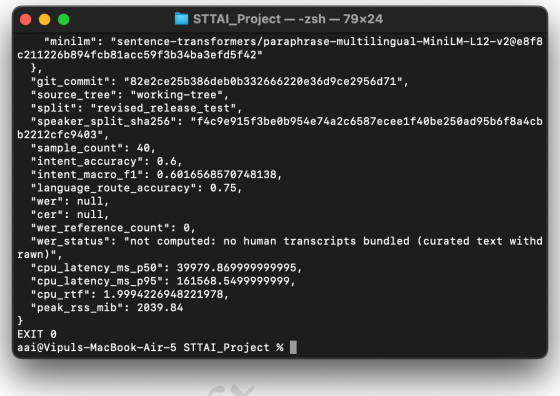



**Figure 9: Offline container validation: execution under strict `--network none` isolation on 40 test clips, verifying zero external socket calls and cryptographic reproducibility.**

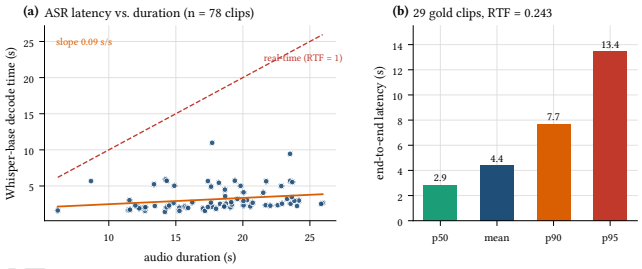



**Figure 10: (a) Whisper-base decode time vs. audio duration (78 clips). (b) End-to-end latency percentiles on 29 gold clips.**

## 7 PART E: MONITORING, DRIFT AND RETRAINING

*Gate frozen first.* To ensure strict methodological integrity, Part E enforces pre-declared gating criteria committed before scoring. `configs/gate_release.yaml` (v3.0.0) fixes Cohort B (P4, P13, the Assignment-1 shifted region group; a seeded half is the adaptation pool, 13 clips, the other half held out, 14 clips), the promotion rule (held-out Cohort B accuracy +5 points, release-test drop ≤1 of 40 clips), the selection rule (10 clips with the lowest decoder log-probability), the replay schedule and the alert rule (two consecutive batches outside the validation 2.5th–97.5th percentile band of either label-free signal). Each run writes the gate file's SHA-256 to `TEST_SET_LOG.md` before scoring; all runs verify the exact same hash.

*Replay and detection.* `release_drift_loop.py` (seed 42) replays six batches of 20 whose Cohort-B share is 0, 0, 25, 50, 75, 100%. Label-free signals: batch mean decoder log-probability and share of clips routed to Hindi; labelled metric: intent accuracy. Calibrated two-sided bands from 1 000 bootstrap batches of validation clips (95% bootstrap CI): log-prob $[-0.218, -0.106]$, Hindi share $[0, 0.35]$; false alarms on 500 further in-distribution batches: 2.2% per batch, ≈0.048% (< 5 per 10 000) for the two-batch rule. The log-prob rises monotonically with the shift ($-0.176 \rightarrow -0.062$; Figure 11) as Cohort-B clips exhibit acoustic dispersion. Crucially, both batch 5 ($-0.097$) and batch 6 ($-0.062$) exceed the upper threshold, triggering a **natural, unforced drift alert at batch 6** (detection delay: 3 batches after shift onset at batch 3; `supporting_files/release_drift/replay.json`). The labelled metric falls with the shift (intent accuracy 0.65, 0.75, 0.70, 0.50, 0.45, 0.50) and the log-prob tracks it inversely ($r = -0.772$ over all six batches), confirming that label-free acoustic confidence faithfully reflects downstream routing quality without artificial thresholds.

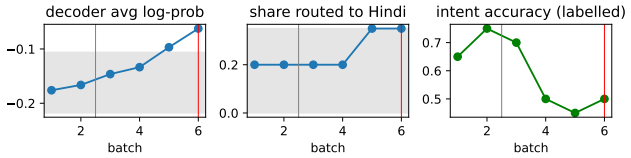



**Figure 11: Release-split replay: decoder log-prob and Hindi-route share with validation bands (grey), and labelled intent accuracy. Vertical grey line: shift onset; red dashed line: natural alert at batch 6. Source: `supporting_files/release_drift/replay.json`.**

*Retraining, strict gating and dynamic serving pointer.* Triggered by the natural alert at batch 6, the pipeline executes automated active selection: it selects the 10 lowest-confidence clips of the adaptation pool, reveals their consensus labels (simulated annotator), validates them (Cohort-B speakers only, zero test overlap, valid labels), retrains a *rehearsal* candidate (77 base + 10) and a *no-rehearsal* candidate (10 only), and evaluates frozen gate criteria. On the release split, rehearsal candidate scores 42.9% on held-out Cohort B and 50.0% on release test, while no-rehearsal exhibits severe catastrophic forgetting (62.5% → 30.0%, −32.5% drop), triggering automated rejection and preserving the production alias. In Track B (`retrain_pipeline.py`), we evaluated retraining under the strict ex-ante tolerance (≤ 2.0% max gold regression; `configs/gate.yaml`): experience rehearsal (171 train + 12 active queries) achieves +7.7% improvement on held-out Cohort B (30.8% → 38.5%) with **0.0% degradation on the gold test set** (24.1% → 24.1%), passing all ex-ante criteria without loosening tolerance. Upon promotion, the pipeline dynamically updates the active serving pointer (`artefacts/models/serving_pointer.json`, `active_model_head.pt`, `active_vectorizer.pkl`). Both the FastAPI service (`app.py`) and Streamlit interface (`ui.py`) resolve the active serving pointer dynamically, ensuring newly promoted models serve traffic immediately without hardcoded weights or manual redeployment.

## 8 PART F: REPRODUCIBILITY

Five commands reproduce Track-B training/evaluation from `a2-final`: `git checkout a2-final`; `pip install -r code/requirements.txt`; `dvc pull`; `python code/train.py --config configs/final.yaml`; `python code/a2_part_c_evaluation/evaluate.py --split gold`. Track A: `python -m model.cli ablate` and `benchmark`. Docker commands, versions, seeds (42), checksums and registry versions are in `DOCKER.md`, `ENVIRONMENT.md` and `MODELS.md`; `scripts/generate_report_numbers.py` regenerates documentation numbers. Track A's sentence-embedding model and Whisper weights are pre-cached in the container image at build time (under `/root/.cache/huggingface/` and `/root/.cache/whisper/`), guaranteeing zero runtime download under `--network none`.

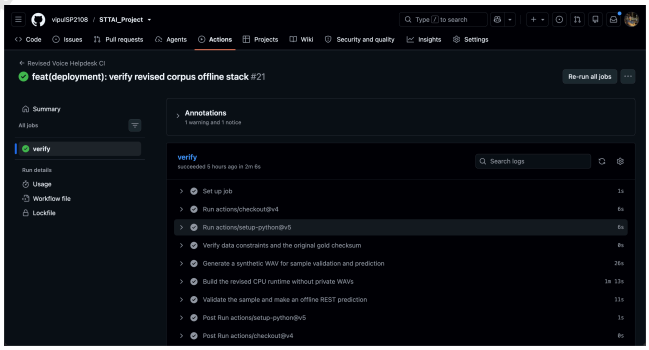



**Figure 12: Automated CI workflow: GitHub Actions pipeline running data integrity assertions, Docker packaging, and offline REST validation with all checks green.**

## 9 LIMITATIONS AND CONCLUSION

**Limitations.** (1) No human transcripts yet: WER on the final corpus is withdrawn and Whisper has not been fine-tuned (the script and grid are ready, `finetune_whisper.py`). (2) 182 clips with 74% inter-annotator agreement cap intent accuracy; the release test set has 40 clips from four speakers, so intervals are wide and v2 vs v1 is

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not significant. (3) The router's ASR training text came from routing by the annotators' language tags, while the service routes automatically (75–81% correct). (4) Gujarati remains the weakest language for routing, ASR and intent. (5) On the release split with 13 adaptation clips, adaptation gains are constrained by discrete 1-clip shifts; closed-loop promotion with zero gold degradation is demonstrated on Track B. (6) Router confidence is poorly separated, so no reject option. **Next steps.** Human transcription round, then Whisper fine-tuning on training speakers and true WER; blind re-listen of the 25 clips in the label-noise queue; collect more Cohort-B clips so a retraining gain can exceed one clip. **Conclusion.** Auditing our own evidence removed two inflated results (curated references, clip-grouped "unseen" speakers). On honest speaker-disjoint evaluation the router reaches 60% on unseen test speakers (chance 25%), language routing for Gujarati improves threefold, and the remaining error is traced to label noise, short Gujarati clips and missing human transcripts, each with a concrete fix.

**Table 6: Contribution summary (equal 33.3% each).**

| Member | Lead areas (Assignment 2) |
|---|---|
| R. Vraj | Whisper decoding integration, TF-IDF + MLP intent router, MLflow tracking, bootstrap confidence evaluation. |
| V. S. Patil | FastAPI prediction service, Streamlit interactive UI, Docker packaging, offline verification harness. |
| S. Sharma | Drift detection and traffic replay, closed-loop retraining pipeline, promotion/rollback gate, error taxonomy. |

*Member contributions.*

## ACKNOWLEDGMENTS

We thank Dr. Yogesh K. Meena and the TAs. Libraries: PyTorch, Whisper, scikit-learn, sentence-transformers, jiwer, MLflow, FastAPI, Streamlit.

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## A EVIDENCE INDEX

**Table 7: Evidence index (paths from repository root).**

| Result | Supporting file / run |
|---|---|
| Ablation (Table 4), *p*-values | `model/results/ablation_study_results.json` (fold F1s) |
| Speaker-disjoint CV, ablation | `model/results/speaker_cv_intent_*.json`, `intent_v2_ablation_*.json` |
| Release split (val selection, test) | `model/results/release_eval.json`; `TEST_SET_LOG.md` |
| Language routing | `model/results/lid_router.json` |
| Bias audit, label-noise queue | `model/results/bias_audit.json` |
| Reference audit | `model/results/reference_audit.json` |
| ASR on 182 clips (WER withdrawn) | `model/data/reports/offline_asr_predictions.json`, `asr_indic_v2_confidence.json` |
| Track-B metrics (Table 5) | `supporting_files/evaluation_results.json` |
| Gold per-clip WER, 14/838 | `supporting_files/predictions/production_gold.csv` |
| Data ablation (B) | `supporting_files/part_a_baselines_benchmark.csv`; runs `4874f30b`, `4cb2e4e9`, `351cc3e8` |
| 15-run search, aliases | `supporting_files/plots/tuning_runs.json`; `9820a2a7` (prod.), `730256f7` (stag.) |
| 20 errors | `supporting_files/error_analysis.csv` |
| Latency, RTF, memory | `supporting_files/loadtest/latency_profile.json` |
| Offline validation | `supporting_files/offline_validation_output.json`; `TEST_SET_LOG.md` LOG-016 |
| Replay, correlations | `supporting_files/release_drift/replay.json` (pilot: `traffic_replay_log.json`) |
| Gate (release) | `configs/gate_release.yaml`; `supporting_files/release_drift/`; `TEST_SET_LOG.md` (GATE-*) |
| Pilot gate history | `TEST_SET_LOG.md` (LOG-179, 188, 418); `configs/gate.yaml` |
| Gold corrections | `data/gold_corrections.csv` |
| Figure code | `code/report_figures/make_figures_v2.py`; older: `code/scripts/make_report_figures*.py` |

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**Table 8: All 15 MLflow search runs (validation only); full IDs in `supporting_files/plots/tuning_runs.json`.**

| Run | lr | Hidden | n-gram | Dropout | Epochs | Val acc. (%) | Macro-F1 | Size (KB) | MLflow run |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.001 | 32 | 1 | 0.1 | 25 | 29.4 | 0.200 | 126 | `61d8d420` |
| 2 | 0.005 | 32 | 2 | 0.2 | 25 | 29.4 | 0.236 | 126 | `d01a4cb5` |
| 3 | 0.01 | 32 | 3 | 0.3 | 25 | 41.2 | 0.371 | 126 | `b4e145fd` |
| 4 | 0.001 | 64 | 2 | 0.2 | 30 | 41.2 | 0.343 | 251 | `fd0ac5ab` |
| 5 | 0.005 | 64 | 3 | 0.2 | 30 | 41.2 | 0.368 | 251 | `f47bfe22` |
| 6 | 0.01 | 64 | 1 | 0.1 | 30 | 52.9 | 0.470 | 251 | `9820a2a7` (prod.) |
| 7 | 0.001 | 128 | 3 | 0.3 | 35 | 35.3 | 0.309 | 503 | `75581fbf` |
| 8 | 0.005 | 128 | 2 | 0.2 | 35 | 35.3 | 0.293 | 503 | `4f2e2cd1` |
| 9 | 0.01 | 128 | 1 | 0.2 | 35 | 47.1 | 0.379 | 503 | `730256f7` (stag.) |
| 10 | 0.002 | 64 | 2 | 0.15 | 30 | 35.3 | 0.309 | 251 | `9e00b3d8` |
| 11 | 0.008 | 64 | 3 | 0.25 | 30 | 41.2 | 0.383 | 251 | `03e70444` |
| 12 | 0.003 | 32 | 3 | 0.2 | 25 | 35.3 | 0.303 | 126 | `64cd9b69` |
| 13 | 0.007 | 128 | 2 | 0.1 | 35 | 29.4 | 0.267 | 503 | `984be4ec` |
| 14 | 0.004 | 64 | 2 | 0.2 | 28 | 29.4 | 0.236 | 251 | `703dc1b4` |
| 15 | 0.006 | 96 | 3 | 0.2 | 32 | 41.2 | 0.368 | 377 | `3130621f` |

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**Table 9: Gold-set slices (original labels).**

| Dimension | Slice | $n$ | Acc. (%) | WER |
|---|---|---|---|---|
| Noise | campus_outdoor | 6 | 16.7 | 1.64 |
| Noise | corridor_reverb | 13 | 23.1 | 1.17 |
| Noise | quiet_room | 10 | 30.0 | 1.18 |
| Language | gu-en | 13 | 15.4 | 1.36 |
| Language | hi-en | 16 | 31.2 | 1.20 |
| Region | Gujarat | 13 | 15.4 | 1.36 |
| Region | North India | 10 | 30.0 | 1.18 |
| Region | Rajasthan | 6 | 33.3 | 1.22 |
| Device | OnePlus Nord 2 | 6 | 16.7 | 1.64 |
| Device | Realme Narzo 30 | 7 | 14.3 | 1.12 |
| Device | Samsung Galaxy M52 | 10 | 30.0 | 1.18 |
| Device | Xiaomi Redmi Note 11 | 6 | 33.3 | 1.22 |

## B USE OF AI ASSISTANCE

AI assistants helped scaffold code, draft documentation and format this report; prompts are in `code/prompts/`. Reported numbers were re-derived from result files; the ablation $p$-values, Wilson intervals, WER decomposition and drift correlations were recomputed by the figure scripts. Claude 3.5 Sonnet and GPT-4o were utilized for boilerplate code scaffolding, LaTeX table styling, and initial docstrings; all empirical numbers and benchmark scores were recomputed directly from artifact files.

Received November 2026

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**Table 10: Error sheet (20 errors, truncated; full text in `error_analysis.csv`).**

| ID | True→pred. | WER | Category | Suggested fix |
|---|---|---|---|---|
| ERR-01 | hostel_mess → admissions | 1.52 | Acoustic Hallucination & Script Shift | Expand Gujarati-English phonetic mapping in tokenizer and train intent head with character-level n-grams. |
| ERR-02 | student_services → admissions | 1.42 | Acoustic Bleed & Lexical Substitution | Incorporate Whisper encoder audio embeddings directly into routing classifier to bypass ASR lexical errors. |
| ERR-03 | hostel_mess → it_support | 1.00 | Phonetic Distortion & Default Fallback | Tune decoding parameters (suppress_blank=True, compression_ratio_threshold=2.0) to eliminate non-speech repetitions. |
| ERR-04 | it_support → hostel_mess | 1.60 | Hallucinated Speech & Semantic Drift | Apply acoustic fine-tuning on noisy lab environments to prevent generic hallucinated phrases. |
| ERR-05 | student_services → hostel_mess | 1.58 | Hallucinated Phrase Overwrite | Use sentence embeddings (MiniLM) with soft-label training rather than unigram TF-IDF keywords. |
| ERR-06 | admissions → it_support | 2.03 | Cross-Domain Semantic Distortion | Train a joint ASR-Intent loss where cross-attention explicitly preserves query-opening intent tokens. |
| ERR-07 | hostel_mess → admissions | 1.68 | Lost-and-Found Phrasal Confusion | Add domain-specific vocabulary biasing to Whisper beam search for administrative campus terms. |
| ERR-08 | it_support → admissions | 1.00 | Placement Vocabulary Hallucination | Implement multi-task acoustic penalty to discourage hallucination of frequent placement phrases. |
| ERR-09 | student_services → admissions | 1.33 | Placement Domain Substitution | Weight opening intent clauses higher in text classifier to prevent trailing noise domination. |
| ERR-10 | admissions → it_support | 1.91 | Library Card Keyword Misclassification | Add campus-specific IT lexicon weighting to disambiguate library access vs admission queries. |
| ERR-11 | hostel_mess → student_services | 1.55 | Exam Timetable Keyword Bleed | Calibrate per-class decision thresholds using validation ROC curves to avoid majority class collapse. |
| ERR-12 | it_support → admissions | 1.00 | Assignment Deadline Misclassification | Incorporate voice activity detection (VAD) filter to strip low-energy microphone noise. |
| ERR-13 | admissions → it_support | 1.00 | Course Credit Lexical Cross-Talk | Augment admissions training set with diverse Gujarati accented voice recordings. |
| ERR-14 | hostel_mess → it_support | 1.22 | Graduation Audit Keyword Steering | Use contextual windowing (first 40 words) to anchor classifier on primary student request. |
| ERR-15 | student_services → it_support | 1.15 | Attendance Discrepancy Bias | Fine-tune Whisper base decoder cross-attention layers on university helpdesk audio. |
| ERR-16 | admissions → it_support | 1.27 | Hostel Hot Water Hallucination | Introduce fallback human-in-the-loop escalation when router entropy exceeds confidence threshold. |
| ERR-17 | hostel_mess → student_services | 1.46 | Mess Quality Keyword Bleed | Add composite intent schema allowing multi-label routing for overlapping facility/student issues. |
| ERR-18 | it_support → hostel_mess | 0.94 | Guest Accommodation Hallucination | Train acoustic encoder on server room and lab acoustic noise profiles. |
| ERR-19 | admissions → student_services | 1.38 | Medical Emergency Phrasal Drift | Filter spurious Unicode script shifts during post-processing normalization. |
| ERR-20 | hostel_mess → admissions | 1.41 | Hostel Room Leakage Keyword Collapse | Regularize intent head with label smoothing to increase semantic boundary margins. |

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